The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 151–175 (PDF)

Papers 655 (concl.), 656, 657, 658, 659, 660, 661 (begin)

Arthur Cayley

Cambridge, 1896

655 (continued): A MEMOIR ON DIFFERENTIAL EQUATIONS.

More symmetrically, we have the solution

$$V = \lambda - \tfrac{1}{2}\left(x^2 + y^2 + z^2\right) + \frac{(ax + by + cz)^2}{a^2 + b^2 + c^2},$$

as can be at once verified.

78. In the same particular case $H = 0$, introducing the corresponding values $V_0, p_0, q_0, r_0, x_0, y_0, z_0$, we find a very simple expression for V_0, as a function of x, y, z, x_0, y_0, z_0. We have, writing $T = e^{\sigma+\sigma'}$, $T_0 = e^{-\sigma+\sigma'}$, and therefore $T_0 T_0' = 1$,

$$\begin{aligned} x - x_0 &= a(T - T_0) + a'(T' - T_0'), \\ y - y_0 &= b(T - T_0) + b'(T' - T_0'), \\ z - z_0 &= c(T - T_0) + c'(T' - T_0'). \end{aligned}$$

Forming the analogous quantities $y - y_0$, &c., we deduce

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = (T - T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2)\,\frac{1}{T^2 T_0^2} \right\},$$

$$(x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2 = (T + T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2)\,\frac{1}{T^2 T_0^2} \right\}.$$

But we have

$$V - V_0 = \tfrac{1}{4}\left\{ (a^2 + b^2 + c^2)(T^2 - T_0^2) + (a'^2 + b'^2 + c'^2)(T'^2 - T_0'^2) \right\},$$

$$T^2 - T_0^2 = \tfrac{1}{4}(T^2 - T_0^2) \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2)\,\frac{1}{T^2 T_0^2} \right\},$$

and hence the required formula

$$V - V_0 = \tfrac{1}{4}\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}\;\sqrt{(x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2},$$

or, say, for shortness,

$$V - V_0 = \tfrac{1}{4}\sqrt{(R)}\,\sqrt{(S)}.$$

79. We ought, therefore, to have

$$\tfrac{1}{4}\,d\left[\sqrt{(R)}\,\sqrt{(S)}\right] = p\,dx + q\,dy + r\,dz - p_0\,dx_0 - q_0\,dy_0 - r_0\,dz_0,$$

where p, q, r, p_0, q_0, r_0 denote as above, and consequently

$$p^2 + q^2 + r^2 = x^2 + y^2 + z^2, \qquad p_0^2 + q_0^2 + r_0^2 = x_0^2 + y_0^2 + z_0^2.$$

We have in fact

$$p = \tfrac{1}{4}\left\{\frac{\sqrt{(R)}}{\sqrt{(S)}}(x - x_0) + \frac{\sqrt{(S)}}{\sqrt{(R)}}(x + x_0)\right\}, \quad \text{\&c.},$$

$$p_0 = \tfrac{1}{4}\left\{-\frac{\sqrt{(R)}}{\sqrt{(S)}}(x - x_0) + \frac{\sqrt{(S)}}{\sqrt{(R)}}(x + x_0)\right\}, \quad \text{\&c.};$$

and thence

$$p^2 + q^2 + r^2 = \tfrac{1}{16}\{R + S + 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2)\} = x^2 + y^2 + z^2,$$

$$p_0^2 + q_0^2 + r_0^2 = \tfrac{1}{16}\{R + S - 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2)\} = x_0^2 + y_0^2 + z_0^2.$$

or the last-mentioned results are thus verified.

Partial Differential Equation containing the Dependent Variable: Reduction to Standard Form.
Art. Nos. 80, 81.

80. The equation $H =$ const. is the most general form of a partial differential equation not containing the dependent variable V; but if a partial differential equation does contain the independent variable, we can, by regarding this as one of the dependent variables, and in place of it introducing a new independent variable, exhibit the equation in the standard form $H =$ const., H being here a homogeneous function of the order zero in the differential coefficients. Thus, if the independent variables are x, y, the dependent variable z, and its differential coefficients p, q, then the given partial differential equation may be $H_1 = (p, q, x, y, z) =$ const. But we may determine z as a function of x, y by an equation $V =$ const., V being a desired function of x, y, z; and then writing p, q, r for the differential coefficients $\dfrac{dV}{dx}, \dfrac{dV}{dy}, \dfrac{dV}{dz}$, we have $p = -\dfrac{p}{r}$, $q = -\dfrac{q}{r}$, and the proposed partial differential equation becomes

$$H = H_1\Big(-\frac{p}{r},\ -\frac{q}{r},\ x,\ y,\ z\Big) = \text{const.}$$

viz. this is an equation containing only the differential coefficients p, q, r of the dependent variable V, a function of x, y, z. And, moreover, H is homogeneous of the order zero in p, q, r; consequently

$$p\,\frac{dH}{dp} + q\,\frac{dH}{dq} + r\,\frac{dH}{dr} = 0,$$

or, in the augmented Hamiltonian system, the last equality is $= -\dfrac{dV}{dt}$, so that an integral is $V =$ const.; as already stated, this is the equation by which z is determined as a function of x, y.

81. Thus, if the given partial differential equation be $pq = z$, we here consider the equation $pq - z = H$. The Hamiltonian system is

$$\frac{r\,dx}{q} = \frac{r\,dy}{p} = \frac{-r^2\,dz}{2pq} = \frac{dp}{0} = \frac{dq}{0} = \frac{dr}{r}\ \left(= -\frac{dV}{0}\right)$$

having the integrals

$$\begin{aligned} a &= p, \\ b &= q, \\ c &= px - qy, \\ d &= \frac{r}{q} - \frac{1}{p}, \\ e &= \frac{x}{pq} - \frac{1}{r^2}, \end{aligned}$$

(whence $H = -abe$). We have H, a, b a system of conjugate integrals and, in terms of these,

$$p = a, \quad q = b, \quad r = \sqrt{(ab/(z+H))};$$

hence, writing λ for the constant value of V, we have

$$\lambda = \int \left\{ a\,dx + b\,dy + \sqrt{\frac{ab}{z+H}}\,dz \right\},$$

that is,

$$\lambda = ax + by + 2\sqrt{[ab(z+H)]},$$

or say,

$$4ab(z+H) = (ax+by-\lambda)^2,$$

a solution containing really the two constants λ and H, and thus a complete solution of the given equation $pq - z = H$. We have, in fact,

$$2ab\,p = a(ax+by-\lambda),$$

$$2ab\,q = b(ax+by-\lambda),$$

that is,

$$4a^2b^2pq = ab(ax+by-\lambda)^2, \; = 4a^2b^2(z+H),$$

or

$$pq = z+H,$$

as it should be.

656.

ON THE THEORY OF PARTIAL DIFFERENTIAL EQUATIONS.

[From the *Mathematische Annalen*, t. XI. (1877), pp. 194–198.]

In what follows, any letter not otherwise explained denotes a function of certain variables (x, y, p, q), or (x, y, z, p, q, r), &c., as will be stated in each particular case.

An equation $a =$ const. denotes that the function a of the variables is, in fact, a constant (viz. by such equation we establish a relation between the variables): and when this is so, we use the same letter a to denote the constant value of the function in question; I find this a very convenient notation.

Thus if the variables are x, y, z, p, q, r and if p, q, r are the differential coefficients in regard to x, y, z respectively of a function V of x, y, z, then H (as a letter not otherwise explained) denotes a function of x, y, z, p, q, r and considering it as a given function,

$$H = \text{const.}$$

will be a partial differential equation containing the constant H. For instance, if H denote the function $pqr - xyz$, $H =$ const. is the partial differential equation, $pqr - xyz =$ const. (a given constant).

The integration of the partial differential equation, $H =$ const., depends upon that of the linear partial differential equation

$$(H, \theta) = 0,$$

where as usual (H, θ) signifies

$$\frac{\partial(H,\theta)}{\partial(p,x)} + \frac{\partial(H,\theta)}{\partial(q,y)} + \frac{\partial(H,\theta)}{\partial(r,z)}.$$

It can be effected if we know two conjugate solutions a, b of the equation $(H, \theta) = 0$, viz. a, b as solutions are such that $(H, a) = 0$, $(H, b) = 0$, and (as conjugate solutions) are also such that $(a, b) = 0$; in this case if from the equations

$$H = \text{const.}, \quad a = \text{const.}, \quad b = \text{const.}$$

we determine p, q, r as functions of x, y, z, the resulting value of $p\,dx + q\,dy + r\,dz$ is an exact differential, and we have

$$V = \lambda + \int (p\,dx + q\,dy + r\,dz),$$

a solution containing three arbitrary constants, λ, a, b, and therefore a complete solution of the proposed partial differential equation $H = \text{const.}$

But (as is known) there is a different process of integration, for which the conjugate solutions are not required, and which has reference to a system of initial values $x_0, y_0, z_0, p_0, q_0, r_0$: viz. if the independent solutions of $(H, \theta) = 0$ are a, b, c, d, e, and if a_0, b_0, c_0, d_0, e_0 denote respectively the same functions of the initial variables that a, b, c, d, e are of x, y, z, p, q, r, then if from the equations

$$a = a_0, \quad b = b_0, \quad c = c_0, \quad d = d_0, \quad e = e_0, \quad H = \text{const.}$$

we express p, q, r as functions of x, y, z and of x_0, y_0, z_0, H, these last being regarded as constants, we have $p\,dx + q\,dy + r\,dz$ an exact differential, and

$$V = \lambda + \int (p\,dx + q\,dy + r\,dz),$$

a solution containing the constants λ, x_0, y_0, z_0 (that is, one supernumerary constant), and as such a complete solution.

It is interesting to prove directly that $p\,dx + q\,dy + r\,dz$ is an exact differential.

I consider first the more simple case where the variables are p, q, x, y. Here p, q are to be found from the equations

$$a = a_0, \quad b = b_0, \quad H = \text{const.}$$

and it is to be shown that $p\,dx + q\,dy$ is an exact differential.

Considering p, q, p_0, q_0 as functions of the independent variables x, y, then differentiating in regard to x, and eliminating $\dfrac{dp_0}{dx}, \dfrac{dq_0}{dx}, \dfrac{dp}{dx}$, we have

$$\begin{vmatrix} \dfrac{da}{dx} + \dfrac{da}{dq}\dfrac{dq}{dx}, & \dfrac{da}{dp}, & \dfrac{da}{dp_0}, & \dfrac{da}{dq_0} \\ \dfrac{db}{dx} + \dfrac{db}{dq}\dfrac{dq}{dx}, & \dfrac{db}{dp}, & \dfrac{db}{dp_0}, & \dfrac{db}{dq_0} \\ \dfrac{dc}{dx} + \dfrac{dc}{dq}\dfrac{dq}{dx}, & \dfrac{dc}{dp}, & \dfrac{dc}{dp_0}, & \dfrac{dc}{dq_0} \\ \dfrac{dH}{dx} + \dfrac{dH}{dq}\dfrac{dq}{dx}, & \dfrac{dH}{dp}, & 0, & 0 \end{vmatrix} = 0,$$

or introducing a well-known notation for functional determinants, and expanding the determinant, this is

$$\frac{\partial(a_0,b_0)}{\partial(p_0,q_0)}\left\{\frac{\partial(H,c)}{\partial(p,x)}+\frac{\partial(H,c)}{\partial(p,q)}\frac{dq}{dx}\right\}+\text{\&c.}=0.$$

But in the same way

$$\frac{\partial(a_0,b_0)}{\partial(p_0,q_0)}\left\{\frac{\partial(H,c)}{\partial(q,y)}+\frac{\partial(H,c)}{\partial(q,p)}\frac{dp}{dy}\right\}+\text{\&c.}=0;$$

or adding these, attending to the value of (H,c), and observing that $\frac{\partial(H,c)}{\partial(p,q)}=-\frac{\partial(H,c)}{\partial(q,p)}$, we have

$$\frac{\partial(a_0,b_0)}{\partial(p_0,q_0)}\left\{(H,c)-\frac{\partial(H,c)}{\partial(p,q)}\left(\frac{dq}{dx}-\frac{dp}{dy}\right)\right\}+\text{\&c.}=0,$$

the terms denoted by the &c. being the like terms with b,c,a and c,a,b in place of a,b,c. We have $(H,a)=0$, $(H,b)=0$, $(H,c)=0$, and the equation in fact is

$$\left\{\frac{\partial(a_0,b_0)}{\partial(p_0,q_0)}\frac{\partial(H,c)}{\partial(p,q)}+\text{\&c.}\right\}\left(\frac{dq}{dx}-\frac{dp}{dy}\right)=0,$$

viz. we have $\frac{dq}{dx}-\frac{dp}{dy}=0$, the condition for the exact differential.

Coming now to the case where the variables are x,y,z,p,q,r, and in the six equations treating p,q,r,p_0,q_0,r_0 as functions of the independent variables x,y,z, then differentiating with regard to x and proceeding as before, we find for $\frac{dr}{dx}$ the equation

$$\frac{\partial(c_0,d_0,e_0)}{\partial(p_0,q_0,r_0)}\left\{\frac{dr}{dx}\frac{\partial(a,b,H)}{\partial(r,p,q)}+\frac{\partial(a,b,H)}{\partial(x,p,q)}\right\}+\text{\&c.}=0.$$

We have, in the same way, for $\frac{dp}{dz}$ the equation

$$\frac{\partial(c_0,d_0,e_0)}{\partial(p_0,q_0,r_0)}\left\{\frac{dp}{dz}\frac{\partial(a,b,H)}{\partial(p,r,q)}+\frac{\partial(a,b,H)}{\partial(z,r,q)}\right\}+\text{\&c.}=0;$$

or, adding the two equations,

$$\frac{\partial(c_0,d_0,e_0)}{\partial(p_0,q_0,r_0)}\left\{\left(\frac{dr}{dx}-\frac{dp}{dz}\right)\frac{\partial(a,b,H)}{\partial(r,p,q)}+\frac{\partial(a,b,H)}{\partial(x,p,q)}-\frac{\partial(a,b,H)}{\partial(z,r,q)}\right\}+\text{\&c.}=0,$$

where the terms denoted by the &c. indicate the like terms corresponding to the different partitions of the letters a,b,c,d,e.

The equation may be simplified; we have identically

$$-H\frac{da_0}{dq_0}\frac{db_0}{dq_0}+H\frac{da_0}{dq_0}\frac{dH_0}{dq_0}+\text{\&c.}=\frac{\partial(a_0,b,H)}{\partial(x,p,q)}-\frac{\partial(a,b,H)}{\partial(z,r,q)},$$

or since $(H, a) = 0$, $(b, H) = 0$, the left-hand side is simply $-\frac{dH}{dq_0}(a, b)$, and the equation becomes

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)}\left\{\left(\frac{dr}{dx} - \frac{dp}{dz}\right)\frac{\partial(a, b, H)}{\partial(r, p, q)} - \frac{dH}{dq_0}(a, b)\right\} + \text{\&c.} = 0.$$

This ought to give $\frac{dr}{dx} - \frac{dp}{dz} = 0$, and we shall in fact have

$$\left\{\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)}\frac{dH}{dq_0}(a, b) + \text{\&c.}\right\} = 0;$$

this is then the equation which has to be proved. By the Poisson-Jacobi theorem, (a, b) is a function of a, b, c, d, e: if we write

$$(a, b) = f(a, b, c, d, e),$$

then (a_0, b_0) is the same function of a_0, b_0, c_0, d_0, e_0; but these are $= a, b, c, d, e$ respectively, and we thence have $(a, b) = (a_0, b_0)$, and the theorem to be proved is

$$\left\{\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)}\frac{dH}{dq_0}(a_0, b_0) + \text{\&c.}\right\} = 0.$$

But substituting for (a_0, b_0) its value, the function on the left-hand is (it is easy to see) the sum of the three functional determinants

$$\frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, q_0, r_0)},\quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, q_0, r_0)},\quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, r_0, q_0)},$$

each of which vanishes as containing the same letter twice in the denominator, that is, as having two identical columns; and the theorem in question is thus proved. And in the same way $\frac{dp}{dy} - \frac{dq}{dx}$, $\frac{dq}{dz} - \frac{dr}{dy}$ are each $= 0$: or we have $p\,dx + q\,dy + r\,dz$ an exact differential.

The proof would fail if the factors multiplying $\frac{dr}{dx} - \frac{dp}{dz}$, &c., or if any one of these factors, were $= 0$; I have not particularly examined this, but the meaning would be, that here the equations in question $a = a_0$, &c., $H =$ const., are such as not to give rise to expressions for p, q, r as functions of $x, y, z, x_0, y_0, z_0, H$, as assumed in the theorem; whenever such expressions are obtainable, then we have $p\,dx + q\,dy + r\,dz$ an exact differential.

The proof in the case of a greater number of variables, say in the next case where the variables are x, y, z, w, p, q, r, s, would present more difficulty; but I have not proceeded further in the question.

It is worth while to put the two processes into connexion with each other: taking in each case the variables to be x, y, z, p, q, r, and the partial differential equation to be $H =$ const.

In the one case, a, b being conjugate solutions of $(H, \theta) = 0$,

from the equations $H = \text{const.}$, $a = \text{const.}$, $b = \text{const.}$,
we find p, q, r functions of x, y, z, H, a, b
and then $p\,dx + q\,dy + r\,dz$ is an exact differential.

In the other case, a, b, c, d, e being the solutions of $(H, \theta) = 0$,

from the equations $H = \text{const.}$, $a = a_0$, $b = b_0$, $c = c_0$, $d = d_0$, $e = e_0$,
we find p, q, r functions of x, y, z, x_0, y_0, z_0, H:
and then $p\,dx + q\,dy + r\,dz$ is an exact differential.

It may be added that, if from the last mentioned equations we determine also p_0, q_0, r_0 as functions of x, y, z, x_0, y_0, z_0, then considering only H as a constant, we ought to have $p\,dx + q\,dy + r\,dz - p_0\,dx_0 - q_0\,dy_0 - r_0\,dz_0$ an exact differential; I have not examined the direct proof.

Cambridge, 28 *Nov.,* 1876.

657.

NOTE ON THE THEORY OF ELLIPTIC INTEGRALS.

[From the *Mathematische Annalen*, t. XII. (1877), pp. 143–146.]

The equation

$$\frac{M\,dy}{\sqrt{1-y^2\,.\,1-k^2y^2}} = \frac{dx}{\sqrt{1-x^2\,.\,1-k^2x^2}}$$

is integrable algebraically when M is rational: and so long as the modulus is arbitrary, then conversely, in order that the equation may be integrable algebraically, M must be rational. For particular values however of the modulus, the equation is integrable algebraically for values of the form M, or (what is the same thing) $\frac{1}{M}$, = a rational quantity + square root of a negative rational quantity, say $= \frac{1}{p}(l + m\sqrt{-n})$, where l, m, n, p are integral and n is positive; we may for shortness call this a half-rational numerical value. The theory is considered by Abel in two Memoirs in the *Astr. Nach.* Nos. 138 & 147 (1828), being the Memoirs[1] XIII & XIV in the Œuvres Complètes (Christiania 1839). I here reproduce the investigation in a somewhat altered (and, as it appears to me, improved) form.

Putting the two differentials each $= du$, we have $x = \operatorname{sn}(u+\alpha)$, $y = \operatorname{sn}\left(\frac{u}{M}+\beta\right)$; and the question is whether there exists an algebraical relation between these functions, or, what is the same thing, an algebraical relation between the functions $x = \operatorname{sn} u$ and $y = \operatorname{sn}\frac{u}{M}$.

Suppose that A and B are independent periods of $\operatorname{sn} u$; so that $\operatorname{sn}(u+A) = \operatorname{sn} u$, $\operatorname{sn}(u+B) = \operatorname{sn} u$, and that every other period is $= mA + nB$, where m and n are integers. Then if u has successively the values $u, u+A, u+2A$, etc., the value of x

[1] They are the Memoirs xix. and xx. in the Œuvres Complètes, t. i., Christiania, 1881.

remains always the same, and if x and y are algebraically connected, y can have only a finite number of values: there are consequently integer values p', p'' for which $\operatorname{sn}\frac{1}{M}(u+p'A) = \operatorname{sn}\frac{1}{M}(u+p''A)$: or writing $u - p'A$ for u and putting $p'' - p' = p$, there is an integer value p for which $\operatorname{sn}\frac{1}{M}(u+pA) = \operatorname{sn}\frac{1}{M}u$.

Similarly there is an integer value q for which $\operatorname{sn}\frac{1}{M}(u+qB) = \operatorname{sn}\frac{1}{M}u$; and we are at liberty to assume $q = p$; for if the original values are unequal, we have only in the place of each of them to substitute their least common multiple.

We have thus an integer p, for which

$$\operatorname{sn}\frac{1}{M}(u+pA) = \operatorname{sn}\frac{1}{M}u,$$

$$\operatorname{sn}\frac{1}{M}(u+pB) = \operatorname{sn}\frac{1}{M}u.$$

There are consequently integers m, n, r, s such that

$$\frac{pA}{M} = mA + nB,$$

$$\frac{pB}{M} = rA + sB,$$

equations which will constitute a single relation $\frac{p}{M} = m$, if $m = s$, $r = n = 0$; but in every other case will be two independent relations. In the case first referred to, the modulus is arbitrary and M is rational.

But excluding this case, the equations give

$$B(mA + nB) = A(rA + sB),$$

or, what is the same thing,

$$rA^2 - (m - s)AB - nB^2 = 0,$$

an equation which implies that the modulus has some one value out of a set of given values. The ratio $A : B$ of the two periods is of necessity imaginary, and hence the integers m, n, r, s must be such that $(m+s)^2 - 4nr$ is negative.

The foregoing equations may be written

$$\left(m - \frac{p}{M}\right) A + nB = 0,$$

$$rA + \left(s - \frac{p}{M}\right) B = 0,$$

whence eliminating A and B we have

$$\left(m - \frac{p}{M}\right)\left(s - \frac{p}{M}\right) - nr = 0,$$

that is,

$$\left(\frac{p}{M}\right)^2 - (m+s)\frac{p}{M} + ms - nr = 0,$$

and consequently

$$\frac{p}{M} = \tfrac{1}{2}(m+s) \pm \tfrac{1}{2}\sqrt{(m-s)^2 + 4nr},$$

where, by what precedes, the integer under the radical sign is negative: and we have thus the above mentioned theorem.

As a very general example, consider the two rational transformations

$$z = z(x, u, v);\ \text{ mod. eq. } Q(u, v) = 0; \quad \frac{N\,dz}{\sqrt{1-z^2\,.\,1-v^4z^2}} = \frac{dx}{\sqrt{1-x^2\,.\,1-u^4x^2}},$$

$$y = y(z, v, w);\ \text{ mod. eq. } P(v, w) = 0; \quad \frac{M\,dy}{\sqrt{1-y^2\,.\,1-w^4y^2}} = \frac{dz}{\sqrt{1-z^2\,.\,1-v^4z^2}};$$

viz. z is taken to be a rational function of x, and of the modular fourth roots u, v; and y to be a rational function of z, and of the modular fourth roots v, w; the transformations being (to fix the ideas) of different orders. We have y a rational function of x, corresponding to the differential relation

$$\frac{MN\,dy}{\sqrt{1-y^2\,.\,1-w^4y^2}} = \frac{dx}{\sqrt{1-x^2\,.\,1-u^4x^2}}.$$

Suppose here $w^4 = u^4$, or say $w = \theta u$, θ being an eighth root of unity: we then have $Q(u, v) = 0$, $P(v, \theta u) = 0$, equations which determine u. The differential equation is then

$$\frac{MN\,dy}{\sqrt{1-y^2\,.\,1-\theta^4u^4y^2}} = \frac{dx}{\sqrt{1-x^2\,.\,1-u^4x^2}},$$

an equation the algebraical integral of which is $y =$ a rational function of x as above; hence, by what precedes, we have $MN =$ a half-rational numerical value, as above.

To explain what the algebraical theorem implied herein is, observe that the equations $Q(u, v) = 0$, $P(v, \theta u) = 0$, give for u an algebraical equation. Admitting θ as an adjoint radical, suppose that an irreducible factor is $\phi(u)$, and take u to be determined by the equation $\phi u = 0$; then v, and consequently also any rational function $\dfrac{1}{MN}$ of u, v, can be expressed as a rational integral function of u, of a degree which is at most equal to the degree of the function ϕu less unity. The theorem is that, in virtue of the equation $\phi u = 0$, this rational function of u becomes equal to a half-rational numerical value as above. Thus in a simple case, which actually presented itself, the equation $\phi u = 0$ was $u^2 - 4u + 1 = 0$; and $\dfrac{1}{MN}$ had the value $u - 2$, which in virtue of this equation becomes $= \pm\sqrt{-3}$.

Thus if the second transformation be the identity $z = y$, $M = v$, $w = v$: we have $v = \theta u$; and the equations are

$$\frac{N\,dy}{\sqrt{1-y^2\,.\,1-\theta^4u^4y^2}} = \frac{dx}{\sqrt{1-x^2\,.\,1-u^4x^2}}, \quad Q(x,u,\theta u) = 0, \quad Q(u,\theta u) = 0.$$

In particular, if the relation between y, x be given by the cubic transformation

$$y = x\,\frac{-v^4x^2 + 2u^2v^2 + v^4}{x^2 - 2u^2v^2x^2 + u^4v^4},$$

so that the modular equation $Q(u,v) = 0$ is $u^4 - v^4 + 2uv(1-u^2v^2) = 0$; then, writing herein $v = \theta u$, and taking θ a prime eighth root of unity, that is, a root of $\theta^4 + 1 = 0$, we have

$$Q(u,\theta u) = -u^4 + \theta^4 + 2\theta u^2(1-\theta^2u^4)$$

viz. disregarding the factor u^2, the equation for u is $u^4 + \theta u^2 + \theta^2 = 0$; or, if ω be an imaginary cube root of unity ($\omega^2 + \omega + 1 = 0$), this is $(u^2 - \omega\theta)(u^2 - \omega^2\theta) = 0$; so that a value of u^2 is $= -\omega\theta$.

Assuming then $\theta^4 + 1 = 0$, $v = \theta u$ and $u^2 = -\omega\theta$, we have $v + 2u^2 = v + \theta u^2(1+2\omega)$, $= -\theta^2\omega(\omega^2 - \omega) := \theta^2\omega'$; $-\omega^2 := -\omega^2, (v + 2u^2)vu^2 = -\omega^2(\omega - \omega^2), u^4 = -\omega^2\theta^2$; and the formula becomes

$$y = x\,\frac{1 - \omega^2(\omega-\omega^2)\theta^2x^2}{1 - \omega(\omega-\omega^2)\theta^2x^2},$$

giving

$$\frac{dy}{\sqrt{1-y^2\,.\,1+\omega\theta^2y^2}} = \frac{(\omega-\omega^2)\,dx}{\sqrt{1-x^2\,.\,1+\omega^2\theta^2x^2}},$$

where as before $\omega^2 + \omega + 1 = 0$, a result which can be at once verified. We have $\omega - \omega^2 = \sqrt{-3}$, or the coefficient $\omega - \omega^2$ in the differential equation is $= \sqrt{-3}$, which is of the form mentioned in the general theorem.

We might, instead of $z = y$, have assumed between y and z the relation corresponding to any other of the six linear transformations of an elliptic integral, and thus have obtained in each case, for a properly determined value of the modulus, a cubic transformation to the same modulus.

Cambridge, 10 *April,* 1877.

658.

ON SOME FORMULÆ IN ELLIPTIC INTEGRALS.

[From the *Mathematische Annalen*, t. XII. (1877), pp. 369–374.]

I reproduce in a modified form an investigation contained in the memoir, Zolotareff, "Sur la méthode d'intégration de M. Tchébychef," *Mathematische Annalen*, t. V. (1872), pp. 560–580.

Starting from the quartic

$$(a, b, c, d, e)(x, 1)^4 = a \cdot \overline{x-\alpha} \,.\, \overline{x-\beta} \,.\, \overline{x-\gamma} \,.\, \overline{x-\delta},$$

we derive from it the quartic

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = a_1 \cdot \overline{x_1-\alpha_1} \,.\, \overline{x_1-\beta_1} \,.\, \overline{x_1-\gamma_1} \,.\, \overline{x_1-\delta_1},$$

where, writing for shortness

$$\lambda = -\alpha + \beta + \gamma - \delta,$$

the roots of the new quartic are

$$\begin{aligned} \alpha_1 &= \theta + \frac{\mu\nu}{2\lambda}, \\ \beta_1 &= \theta + \frac{\nu\lambda}{2\mu}, \\ \gamma_1 &= \theta + \frac{\lambda\mu}{2\nu}, \\ \delta_1 &= \theta, \end{aligned}$$

where

$$\mu = \alpha - \beta + \gamma - \delta, \quad \nu = \alpha + \beta - \gamma - \delta,$$

θ being arbitrary: the differences of the roots $\alpha_1, \beta_1, \gamma_1, \delta_1$ are, it will be observed, functions of the differences of the roots $\alpha, \beta, \gamma, \delta$.

We assume $a_1 = a = 1$, nevertheless retaining in the formulae a, or a_1 (each meaning 1), whenever, for the sake of homogeneity, it is convenient to do so. The relations between the remaining coefficients b_1, c_1, d_1, e_1, and b, c, d, e, are of course to be calculated from the formulae $-4b = \Sigma\alpha$, $6c = \Sigma\alpha\beta$, &c., and the like formulae $-4b_1 = \Sigma\alpha_1$, $6c_1 = \Sigma\alpha_1\beta_1$, &c. We thus have

$$-4b_1 = 4\theta + \tfrac{1}{2}\Sigma\frac{\mu\nu}{\lambda},$$

$$-4d_1 = 4\theta^3 + \tfrac{3}{2}\theta^2\Sigma\frac{\mu\nu}{\lambda} + \tfrac{1}{4}\theta\Sigma\lambda^2 + \tfrac{1}{8}\lambda\mu\nu,$$

where $\Sigma\dfrac{\mu\nu}{\lambda} = \dfrac{1}{\lambda\mu\nu}\Sigma\lambda^2\mu^2$.

Writing, for shortness,

$$\begin{aligned}
C &= ac - b^2,\\
D &= a^2d - 3abc + 2b^3,\\
E &= a^3e - 4a^2bd + 6ab^2c - 3b^4 = a^2I - 3C^2,\\
I &= ae - 4bd + 3c^2,\\
J &= ace - ad^2 - b^2e + 2bcd - c^3,
\end{aligned}$$

we have

$$a^2E = -a^2I + 12C^2,$$

$$\Sigma\lambda^2 = -4(b + B),$$

$$\Sigma\lambda^4 = -48C,$$

$$\Sigma\lambda^2\mu^2 = 24a^2c - 8(b + B)^2,$$

$$\Sigma\lambda\mu\nu = 32D,$$

$$\Sigma\lambda^2\mu^2\nu^2 = 64(-a^2I + 12C^2),$$

where the last equation may be verified by means of the formula

$$(\Sigma\lambda^2)^2 = \Sigma\lambda^4 + 2\Sigma\lambda^2\mu^2.$$

And we hence obtain

$$\begin{aligned}
a_1 &= 1,\\
b_1 &= -\theta - B,\\
c_1 &= \theta^2 + 2B\theta - 2C,\\
d_1 &= -\theta^3 - 3B\theta^2 + 6C\theta - D,\\
e_1 &= \theta^4 + 4B\theta^3 - 12C\theta^2 + 4D\theta.
\end{aligned}$$

And consequently

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = (1, -B, -2C, -D, 0)(x_1 - \theta, 1)^4.$$

Hence also

$$\begin{aligned} I_1 &= a_1 e_1 - 4b_1 d_1 + 3c_1^2 = -4BD + 12C^2 = a^2 I, \\ J_1 &= a_1 c_1 e_1 - a_1 d_1^2 - b_1^2 e_1 + 2b_1 c_1 d_1 - c_1^3 = -D^2 + 8C^3 - 4BCD \\ &= -D^2 + 8C^3 + C(a^2 I - 12C^2) \\ &= a^2 CI - 4C^3 - D^2 \\ &= a^2 J; \end{aligned}$$

where, as regards this last equation $a^2CI - 4C^3 - D^2 = a^2J$, observe that C and D are the leading coefficients of the Hessian H and the cubicovariant Φ of the quartic function U, and hence that the identity $-\Phi^2 = JU^3 - IU^2H + 4H^3$, attending only to the term in a^6, becomes $-D^2 = a^4J - a^2CI + 4C^3$, which is the equation in question.

We thus have $I_1 = I$, $J_1 = J$; viz. the functions $(a, b, c, d, e)(x, 1)^4$, $(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4$, are linearly transformable the one into the other, and that by a unimodular substitution $x_1 = \rho x + \sigma$, $x = \rho' x_1 + \sigma'$, where $\rho\sigma' - \rho'\sigma = 1$. It may be remarked that we have $(a, b, c, d, e)(x, 1)^4 = (1, 0, C, D, E)(x + b, 1)^4$; and hence the theorem may be stated in the form: the quartic functions $(1, 0, C, D, E)(x, 1)^4$, and $(1, -B, -2C, -D, 0)(x_1, 1)^4$ are transformable the one into the other by a unimodular substitution: or again, substituting for E its value $a^2I - 3C^2$, $= -4BD + 9C^2$, the quartic functions

$$(1, 0, C, D, -4BD + 9C^2)(x, 1)^4, \text{ and } (1, -B, -2C, -D, 0)(x_1, 1)^4$$

are linearly transformable the one into the other by a unimodular substitution. In this last form B, C, D are arbitrary quantities; it is at once verified that the invariants I, J have the same values for the two functions respectively; and the theorem is thus self-evident.

Reverting to the expressions for $\alpha_1, \beta_1, \gamma_1, \delta_1$ we obtain

$$\begin{aligned} \alpha_1 - \delta_1 &= \frac{\mu\nu}{2\lambda} = \tfrac{1}{2\lambda}(\alpha - \delta)(\beta - \gamma), \\ \beta_1 - \delta_1 &= \frac{\nu\lambda}{2\mu} = \tfrac{1}{2\mu}(\beta - \delta)(\gamma - \alpha), \\ \gamma_1 - \delta_1 &= \frac{\lambda\mu}{2\nu} = \tfrac{1}{2\nu}(\gamma - \delta)(\alpha - \beta), \end{aligned}$$

$$\beta_1 - \gamma_1 = \tfrac{1}{2\lambda}(\overline{\alpha - \delta} \,.\, \overline{\beta - \gamma}), \quad \gamma_1 - \alpha_1 = \tfrac{1}{2\mu}(\overline{\beta - \delta} \,.\, \overline{\gamma - \alpha}), \quad \alpha_1 - \beta_1 = \tfrac{1}{2\nu}(\overline{\gamma - \delta} \,.\, \overline{\alpha - \beta}).$$

Hence also

$$\begin{aligned} &\overline{\alpha - \delta} \,.\, \overline{\beta - \gamma} \,.\, \overline{\beta - \delta} \,.\, \overline{\gamma - \alpha} \,.\, \overline{\gamma - \delta} \,.\, \overline{\alpha - \beta} \\ &= \overline{\alpha_1 - \delta_1} \,.\, \overline{\beta_1 - \gamma_1} \,.\, \overline{\beta_1 - \delta_1} \,.\, \overline{\gamma_1 - \alpha_1} \,.\, \overline{\gamma_1 - \delta_1} \,.\, \overline{\alpha_1 - \beta_1}, \end{aligned}$$

which agrees with the foregoing equations $I_1 = I$ and $J_1 = J$, since I, J are functions of the first set of quantities and I_1, J_1 the like functions of the second set; in fact, $I = \frac{1}{3}(P^2 + Q^2 + R^2)$, and $J = \frac{1}{27}(Q - R)(R - P)(P - Q)$, if for a moment the quantities are called P, Q, R.

We consider now the differential expression $\dfrac{dx}{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}}}$; to transform this into the elliptic form, assume

$$k^2 = \frac{\alpha-\beta}{\gamma-\alpha}\cdot\frac{\gamma-\delta}{\beta-\delta},\quad \operatorname{sn}^2\alpha = \frac{\gamma-\alpha}{\gamma-\delta},$$

(where α is of course not the coefficient, $= 1$, heretofore represented by that letter: as α will only occur under the functional signs sn, cn, dn, there is no risk of ambiguity). And then further

$$x = \frac{-\alpha\operatorname{sn}^2 u - \delta\operatorname{sn}^2\alpha - u}{\operatorname{sn}^2\alpha - \operatorname{sn}^2 u}.$$

Forming the equations

$$x-\alpha = \frac{\alpha-\beta}{\gamma-\alpha}\cdot\&\text{c.},\quad \&\text{c.},$$

we deduce without difficulty

$$\operatorname{sn}^2\alpha = \frac{\gamma-\alpha}{\gamma-\delta},\quad \frac{\operatorname{sn}^2 u}{\operatorname{sn}^2\alpha} = \frac{x-\delta}{x-\alpha},$$

$$\operatorname{cn}^2\alpha = \frac{\alpha-\delta}{\gamma-\delta},\quad \frac{\operatorname{cn}^2 u}{\operatorname{cn}^2\alpha} = \frac{x-\gamma}{x-\alpha},$$

$$\operatorname{dn}^2\alpha = \frac{\alpha-\beta}{\gamma-\beta},\quad \frac{\operatorname{dn}^2 u}{\operatorname{dn}^2\alpha} = \frac{x-\beta}{x-\alpha},$$

$$1-k^2\operatorname{sn}^4\alpha = \frac{(\alpha-\delta)(-\alpha+\beta+\gamma-\delta)}{\overline{\beta-\delta}\,.\,\overline{\gamma-\delta}} = \frac{\lambda(\alpha-\delta)}{\overline{\beta-\delta}\,.\,\overline{\gamma-\delta}},$$

the use of which last equation will presently appear.

We hence obtain

$$2\operatorname{sn} u\operatorname{cn} u\operatorname{dn} u\,du = -(\alpha-\delta)\operatorname{sn}^2\alpha\,\frac{dx}{(x-\alpha)^2},$$

$$\operatorname{sn} u\operatorname{cn} u\operatorname{dn} u = \frac{1}{\operatorname{sn}\alpha\operatorname{cn}\alpha\operatorname{dn}\alpha}\,\frac{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}}}{(x-\alpha)^2},$$

and consequently

$$2\,du = -\frac{(\alpha-\delta)\operatorname{sn}\alpha}{\operatorname{cn}\alpha\operatorname{dn}\alpha}\,\frac{dx}{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}}},$$

or, reducing the coefficient,

$$\frac{dx}{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}}} = \frac{-2}{\sqrt{\overline{\gamma-\alpha}\,.\,\overline{\beta-\delta}}}\,du,$$

which is the required formula.

We next have

$$k^2 = \frac{4\operatorname{sn}^2\alpha\operatorname{cn}^2\alpha\operatorname{dn}^2\alpha}{4} = \frac{\overline{\beta-\delta}\,.\,\overline{\gamma-\alpha}}{\gamma_1-\alpha_1}\cdot\quad = \&\text{c.}$$

in virtue of the foregoing values

$$\gamma_1 - \alpha_1 = \tfrac{1}{2\mu}(\beta-\delta)(\gamma-\alpha) \text{ and } \gamma_1 - \delta_1 = \frac{\lambda\mu}{2\nu}.$$

Moreover

$$k_1^2 = -\frac{\overline{\alpha-\beta}\,.\,\overline{\gamma-\delta}}{\overline{\gamma-\alpha}\,.\,\overline{\beta-\delta}} = \frac{\overline{\alpha_1-\beta_1}\,.\,\overline{\gamma_1-\delta_1}}{\overline{\gamma_1-\alpha_1}\,.\,\overline{\beta_1-\delta_1}}.$$

Hence the like formulæ with the same value of k^2, and with 2α in place of α, will be applicable to the like differential expression in x_1: viz. assuming

$$x_1 = \frac{-\alpha_1 \operatorname{sn}^2 u_1 - \delta_1 \operatorname{sn}^2 2\alpha}{\operatorname{sn}^2 u_1 - \operatorname{sn}^2 2\alpha},$$

we have

$$\frac{dx_1}{\sqrt{\overline{x_1-\alpha_1}\,.\,\overline{x_1-\beta_1}\,.\,\overline{x_1-\gamma_1}\,.\,\overline{x_1-\delta_1}}} = \frac{-2}{\sqrt{\overline{\gamma_1-\alpha_1}\,.\,\overline{\beta_1-\delta_1}}}\,du_1.$$

We have thus the integral of the differential equation

$$\frac{dx_1}{\sqrt{\overline{x_1-\alpha_1}\,.\,\overline{x_1-\beta_1}\,.\,\overline{x_1-\gamma_1}\,.\,\overline{x_1-\delta_1}}} = \frac{dx}{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}}}$$

(the two quartic functions being of course connected as before); viz. assuming x, x_1 functions of u, u_1 respectively as above and recollecting that $\overline{\gamma_1-\alpha_1}\,.\,\overline{\beta_1-\delta_1} = \overline{\gamma-\alpha}\,.\,\overline{\beta-\delta}$,

$du_1 = du$; and therefore $u_1 = u + f$ (f an arbitrary constant); the required

integral is thus given by the equations

$$\frac{x-\delta}{x-\alpha} = \frac{\operatorname{sn}^2 u}{\operatorname{sn}^2 \alpha}, \quad \frac{x_1-\delta_1}{x_1-\alpha_1} = \frac{\operatorname{sn}^2(u+f)}{\operatorname{sn}^2 2\alpha}, \quad (f \text{ constant of integration}).$$

Using the formula

$$\operatorname{sn}^2(u+f) - \operatorname{sn}^2 f = \frac{\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u \operatorname{sn} f \operatorname{cn} f \operatorname{dn} f}{1}$$

we obtain

$$\{(x-\delta)\operatorname{sn}^2\alpha - (x-\alpha)\operatorname{sn}^2 f\}^2 \times \frac{x_1-\delta_1}{\operatorname{sn}^2 2\alpha} = \{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\delta}} \operatorname{sn}\alpha \operatorname{cn} f \operatorname{dn} f - \sqrt{\overline{x-\beta}\,.\,\overline{x-\gamma}} \operatorname{sn} f \operatorname{cn}\alpha \operatorname{dn}\alpha\}^2,$$

which is the general integral.

We obtain a particular integral of a very simple form by assuming $f = \alpha$, viz. this is

$$\frac{x_1-\alpha_1}{x_1-\delta_1}\,\frac{}{\operatorname{sn}^2 2\alpha} = \operatorname{cn}^2\alpha \operatorname{dn}^2\alpha\,\{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\delta}} - \sqrt{\overline{x-\beta}\,.\,\overline{x-\gamma}}\}^2,$$

this is

$$\frac{x_1-\delta_1\,.\,\gamma_1-\beta_1}{x_1-\alpha_1\,.\,\gamma_1-\delta_1} = \frac{\overline{\gamma-\alpha}\,.\,\overline{\beta-\delta}}{\{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\delta}} - \sqrt{\overline{x-\beta}\,.\,\overline{x-\gamma}}\}^2}.$$

or writing $\overline{\gamma-\alpha}\,.\,\overline{\beta-\delta}=\overline{\gamma_1-\alpha_1}\,.\,\overline{\beta_1-\delta_1}$, reducing and inverting, we have

$$\frac{\alpha_1-\beta_1\,.\,\beta_1-\delta_1}{x_1-\delta_1}=\frac{1}{\beta_1-\delta_1\,.\,\alpha_1-\delta_1}\{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\delta}}-\sqrt{\overline{x-\beta}\,.\,\overline{x-\gamma}}\}^2.$$

which may also be written in the equivalent forms

$$\frac{x_1-\beta_1}{x_1-\alpha_1\,.\,\beta_1-\delta_1}=\tfrac{1}{}\{\sqrt{\overline{x-\gamma}\,.\,\overline{x-\delta}}-\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}}\}^2,$$

$$\frac{x_1-\delta_1}{x_1-\beta_1\,.\,\alpha_1-\delta_1}=\tfrac{1}{}\{\sqrt{\overline{x-\gamma}\,.\,\overline{x-\delta}}-\sqrt{\overline{x-\alpha}\,.\,\overline{x-\beta}}\}^2.$$

In fact, from the first equation we have

$$\frac{x_1-\alpha_1\,.\,\beta_1-\delta_1\,.\,\gamma_1-\delta_1}{x_1-\delta_1}=(x-\alpha)(x-\delta)-\{\sqrt{\overline{x-\alpha}\,.\,\overline{x-\delta}}-\sqrt{\overline{x-\beta}\,.\,\overline{x-\gamma}}\}^2,$$

where the expression on the right-hand side is

$$\delta_1^2-\delta_1(\alpha_1+\beta_1+\gamma_1)+\alpha_1\beta_1+\beta_1\gamma_1-2\alpha_1+\delta_1^2(\alpha+\beta+\gamma+\delta)-\alpha\delta-\beta\gamma+2\sqrt{X},$$

X having here the value

$$X=\overline{x-\alpha}\,.\,\overline{x-\beta}\,.\,\overline{x-\gamma}\,.\,\overline{x-\delta}.$$

Writing for a moment

$$P=\alpha\delta+\beta\gamma,\quad P_1=\alpha_1\delta_1+\beta_1\gamma_1,$$
$$Q=\beta\delta+\gamma\alpha,\quad Q_1=\beta_1\delta_1+\gamma_1\alpha_1,$$
$$R=\gamma\delta+\alpha\beta,\quad R_1=\gamma_1\delta_1+\alpha_1\beta_1,$$

then, by what precedes, Q_1-R_1, R_1-P_1, P_1-Q_1 are equal to $Q-R$, $R-P$, $P-Q$ respectively; that is, $P_1-P=Q_1-Q=R_1-R=$ (suppose) Ω, a function symmetrical in regard to $\alpha_1,\beta_1,\gamma_1;\alpha,\beta,\gamma$: the equation therefore is

$$\frac{x_1-\alpha_1\,.\,\beta_1-\delta_1\,.\,\gamma_1-\delta_1}{x_1-\delta_1}=\alpha_1(\beta_1-\alpha_1-\beta_1-\gamma_1)-2\alpha+\delta_1(\alpha+\beta+\gamma+\delta)+2\sqrt{X}+\Omega,$$

or the relation is symmetrical in regard to $\alpha_1,\beta_1,\gamma_1;\alpha,\beta,\gamma$: and the first form implies therefore each of the other two forms.

Cambridge, 8 *May,* 1877.

659.

A THEOREM ON GROUPS.

[From the *Mathematische Annalen*, t. XIII. (1878), pp. 561–565.]

The following theorem is very simple; but it seems to belong to a class of theorems, the investigation of which is desirable.

I consider a substitution-group of a given order upon a given number of letters; and I seek to double the group, that is to derive from it a group of twice the order upon twice the number of letters. This can be effected for any group, in a manner which is self-evident and in nowise interesting: but in a different manner for a commutative group (or group such that any two of its substitutions satisfy the condition $AB = BA$): it is to be observed that the double group is not in general commutative.

Let the letters of the original group be $abcde\ldots$, we may for shortness write $U = abcde\ldots$; and take U as the primitive arrangement: and let the group then be $1, A, B, \ldots$ where $A, B, \ldots$ represent substitutions: the corresponding arrangements are $U, AU, BU, \ldots$ and these may for shortness be represented by $1, A, B, \ldots$; viz. $1, A, B, \ldots$ represent, properly and in the first instance, substitutions; but when it is explained that they represent arrangements, then they represent the arrangements $U, AU, BU, \ldots$.

For the double group the letters are taken to be $a_1b_1c_1d_1e_1\ldots$ and $a_2b_2c_2d_2e_2\ldots = U_1$ and U_2 suppose, and U_1U_2 is regarded as the primitive arrangement; A_1 and A_2 denote the same substitutions in regard to U_1 and U_2 respectively, that A denotes in regard to U: and so for B_1, B_2, etc.; moreover 12 denotes the substitution $(a_1a_2)(b_1b_2)(c_1c_2)(d_1d_2)(e_1e_2)\ldots$, or interchange of the suffixes 1 and 2. The substitutions A_1^i, A_2^j, or any powers of these A_1^i, A_2^j, are obviously commutative; applying them to the primitive arrangement U_1U_2, we have $A_1^iA_2^jU_1U_2$ and $A_2^jA_1^iU_1U_2$, each $= A_1^iU_1\,A_2^jU_2$. But A_1^i, A_2^j are not commutative with 12: we have for instance $12A_1^i\,.\,U_1U_2 = 12A_1^iU_1\,.\,U_2 = A_2^iU_2\,.\,U_1$ but $A_1^i\,12\,U_1U_2 = A_1^i\,.\,U_2U_1 = U_2\,.\,A_1^iU_1$. If instead of the substitutions we consider the arrangements obtained by operating upon U_1U_2, then we may for shortness consider for instance A_1A_2 as denoting the arrangement $A_1U_1 . A_2U_2$. But observe that in this use of the symbols the A_1, A_2 are not commutative, A_2A_1 would denote the different arrangement $A_1U_2 . A_2U_1$: in this use of the symbols, 1 would denote U_1U_2, and 12 would denote U_2U_1, but it would be clearer to use $12, 21$ as denoting U_1U_2 and U_2U_1 respectively.

These explanations having been given, I remark that in every case the substitution-group $1, A, B, \ldots$ gives the double group

$$\begin{aligned}&1,\ \ A_1A_2,\ \ B_1B_2, \ldots\\ &12,\ \ 12A_1A_2,\ \ 12B_1B_2, \ldots\end{aligned}$$

as is at once seen to be true: but further when the original group $1, A, B, \ldots$ is commutative, then if m be any integer number, such that $m^2 = 1$ (mod the order of the original group), we have also the double group

$$\begin{aligned}&1,\ \ A_1A_2^m,\ \ B_1B_2^m, \ldots\\ &12,\ \ 12A_1A_2^m,\ \ 12B_1B_2^m, \ldots\end{aligned}$$

where of course if the order of the original group ($= \mu$ suppose) be prime, we have $m^2 = 1$ (mod μ), say $m = 1$ or else $m = \mu - 1$; but if the order μ be composite, then the number of solutions may be greater.

The condition in order to the existence of the double group of course is that, in the system of substitutions just written down, the combination of any two substitutions may give a substitution of the system. And this is in fact the case in virtue of the formulae

$$\begin{aligned}&1^\circ.\quad A_1A_2^m . B_1B_2^m = A_1B_1(A_2B_2)^m,\\ &2^\circ.\quad A_1A_2^m . 12B_1B_2^m = 12A_1^mB_1(A_2^mB_2)^m,\\ &3^\circ.\quad 12A_1A_2^m . B_1B_2^m = 12(A_1B_2)(A_2B_1)^m,\\ &4^\circ.\quad 12A_1A_2^m . 12B_1B_2^m = A_1^mB_1(A_2^mB_2)^m,\end{aligned}$$

inasmuch as $1, A, B, \ldots$ being a group, AB and A^mB are each of them a substitution of the group, $= C$ suppose; we have of course in like manner $A_1B_1 = C_1$, $A_2B_2 = C_2$, etc., and the right-hand sides of the four formulæ are thus of the forms $C_1C_2^m$, $12C_1C_2^m$, $12C_1C_2^m$, $C_1C_2^m$ respectively, viz. these are substitutions of the system.

To prove for instance the formula 2°, considering the arrangements obtained by operating upon U_1U_2, we have

$$B_1B_2^mU_1U_2 = B_1B_2^m,\ \ 12B_1B_2^mU_1U_2 = B_2^mB_1,\ \ A_1A_2^m . 12B_1B_2^mU_1U_2 = A_1B_2^m . A_2^mB_1.$$

where of course the expressions on the right-hand side denote arrangements. By reason that the original group is commutative $(A^{m-1}B)^m$ is $= A^{m^2-m}B^m$ or since $m^2 = 1$ (mod μ) this is $= A^{1-m}B^m$; hence also $(A_2^{m-1}B_2)^m = A_2^{1-m}B_2^m$: hence, considering as before the arrangements obtained by operating on U_1U_2, we have

$$(A_2^{m-1}B_2)^m U_1U_2 = 1\,.\,A_2B_2^m;\quad A_1^mB_1(A_2^{m-1}B_2)^mU_1U_2 = A_1^mB_1A_2B_2^m,$$

and

$$12A_1^mB_1(A_2^{m-1}B_2)^mU_1U_2 = A_2^mB_2A_1B_1^m,$$

where of course the right-hand sides denote arrangements. Hence in the formula 2°, the two substitutions operating on U_1U_2 give each of them the same arrangement $A_2^mB_2A_1B_1^m$, that is, the two substitutions are equal. And similarly the other formulae $1°, 3°, 4°$ may be proved.

By interchanging A and B, in the formulae I obtain

$$1°.\quad A_1A_2^m\,.\,B_1B_2^m = A_1B_1(A_2B_2)^m;$$

$$B_1B_2^m\,.\,A_1A_2^m = B_1A_1(B_2A_2)^m = A_1B_1(A_2B_2)^m,$$

which is

$$= A_1A_2^m\,.\,B_1B_2^m.$$

$$2° \text{ and } 3°.\quad A_1A_2^m\,.\,12B_1B_2^m = 12A_1^mB_1(A_2^{m-1}B_2)^m;$$

$$12B_1B_2^m\,.\,A_1A_2^m = 12B_1A_1(B_2A_2)^m = 12A_1B_1(A_2B_2)^m,$$

which is not

$$= A_1A_2^m\,.\,12B_1B_2^m.$$

$$3° \text{ and } 2°.\quad 12A_1A_2^m\,.\,B_1B_2^m = 12A_1B_2(A_2B_1)^m;$$

$$B_1B_2^m\,.\,12A_1A_2^m = 12A_1B_2^m(A_2B_1^m)^m = 12A_1B_2^m\,A_2^mB_1,$$

which is not

$$= 12A_1A_2^m\,.\,B_1B_2^m.$$

$$4°.\quad 12A_1A_2^m\,.\,12B_1B_2^m = A_1^mB_1(A_2^{m-1}B_2)^m;$$

which is not

$$12B_1B_2^m\,.\,12A_1A_2^m = A_1B_1^m(A_2B_2^m)^m = (A_1^mB_1)^mA_2^mB_2,$$

$$= 12A_1A_2^m\,.\,12B_1B_2^m.$$

That is, in the double group any two substitutions of the form $A_1A_2^m$ are commutative, but a substitution of this form is not in general commutative with a substitution of the form $12B_1B_2^m$, nor are two substitutions of the last-mentioned form $12A_1A_2^m$ in general commutative with each other; hence the double group is not in general commutative.

In the formula 4°, writing $B = A$, we have

$$(12A_1A_2^m)^2 = A_1^{m+1}A_2^{m^2+m} = A_1^{m+1}A_2^{m+1},$$

hence, if λ is the least integer value such that

$$\lambda(m+1) \equiv 0 \pmod{\mu},$$

we have $(12A_1A_2^m)^\lambda = 1$, viz. in the double group the substitutions of the second row are each of them of an order not exceeding 2λ, the substitution 12 being of course of the order 2. In particular, if $m = \mu - 1$, then $\lambda = 1$: and the substitutions of the second row are each of them of the order 2.

As the most simple instance of the theorem, suppose that the original group is the group $1, (abc), (acb)$, or say $1, \Theta, \Theta^2$, of the cyclical substitutions upon the 3 letters abc. Here $m^2 = 1$ (mod 3) or except $m = 1$ the only solution is $m = 2$, and thence $\lambda = 1$. The double group is a group of the order 6 on the letters $a_1b_1c_1a_2b_2c_2$: viz. writing $\Theta = (abc)$, and therefore $\Theta_1 = (a_1b_1c_1)$, $\Theta_1^2 = (a_1c_1b_1)$, $\Theta_2 = (a_2b_2c_2)$, $\Theta_2^2 = (a_2c_2b_2)$, also writing $12 = \alpha$, the substitutions are

$$\begin{array}{l} 1,\ \ \Theta_1\Theta_2^2,\ \ \Theta_1^2\Theta_2, \\ \alpha,\ \ \alpha\Theta_1\Theta_2^2,\ \ \alpha\Theta_1^2\Theta_2, \end{array}$$

the arrangements corresponding to the second row of substitutions are $a_2b_2c_2a_1b_1c_1$, $b_2c_2a_2c_1a_1b_1$, $c_2a_2b_2b_1c_1a_1$, viz. the substitutions are $(a_1a_2)(b_1b_2)(c_1c_2)$, $(a_1b_2)(b_1c_2)(c_1a_2)$, $(a_1c_2)(b_1a_2)(c_1b_2)$, each of them of the second order as they should be.

I take the opportunity of mentioning a further theorem. Let μ be the order of the group, and α the order of any term A thereof, α being of course a submultiple of μ: and let the term A be called quasi-positive when $\mu\left(1 - \dfrac{1}{\alpha}\right)$ is even, quasi-negative when $\mu\left(1 - \dfrac{1}{\alpha}\right)$ is odd. The theorem is that the product of two quasi-positive terms, or of two quasi-negative terms, is quasi-positive; but the product of a quasi-positive term and a quasi-negative term is quasi-negative. And it follows hence that, either the terms of a group are all quasi-positive, or else one half of them are quasi-positive and the other half of them are quasi-negative.

The proof is very simple: a term A of the group operating on the μ terms $(1, A, B, C, \ldots)$ of the group, gives these same terms in a different order, or it may be regarded as a substitution upon the μ symbols $1, A, B, C, \ldots$; so regarded it is a regular substitution (this is a fundamental theorem, which I do not stop to prove), and hence since it must be of the order α it is a substitution composed of $\dfrac{\mu}{\alpha}$ cycles, each of α letters. But in general a substitution is positive or negative according as it is equivalent to an even or an odd number of inversions; a cyclical substitution upon α letters is positive or negative according as $\alpha - 1$ is even or odd; and the substitution composed of the $\dfrac{\mu}{\alpha}$ cycles is positive or negative according as $\dfrac{\mu}{\alpha}(\alpha - 1)$, that is, $\mu\left(1 - \dfrac{1}{\alpha}\right)$, is even or odd. Hence by the foregoing definition, the term A, according as it is quasi-positive or quasi-negative, corresponds to a positive substitution or to a negative substitution; and such terms combine together in like manner with positive and negative substitutions.

Cambridge, 3rd April, 1878.

660.

ON THE CORRESPONDENCE OF HOMOGRAPHIES AND ROTATIONS.

[From the *Mathematische Annalen*, t. XV. (1879), pp. 238–240.]

It is a fundamental notion in Prof. Klein's theory of the "Icosahedron" that homographies correspond to rotations (of a solid body about a fixed point): in such wise that, considering the homographies which correspond to two given rotations, the homography compounded of these corresponds to the rotation compounded of the two rotations.

Say the two homographies are $A + Bp + Cq + Dpq = 0$, $A_1 + B_1q + C_1r + D_1qr = 0$, then, eliminating q, the compound homography is $A_2 + B_2p + C_2r + D_2pr = 0$, where

$$A_2, B_2, C_2, D_2 = B_1A - A_1C,\ \ B_1B - A_1D,\ \ D_1A - C_1C,\ \ D_1B - C_1D;$$

and the theorem is that, corresponding to these, we have rotations depending on the parameters (λ, μ, ν), $(\lambda_1, \mu_1, \nu_1)$, $(\lambda_2, \mu_2, \nu_2)$ respectively, such that the third rotation is that compounded of the first and second rotations. The question arises to find the expression for the parameters of the homography in terms of the parameters of the corresponding rotation.

The rotation (λ, μ, ν) is taken to denote a rotation through an angle ϑ about an axis the inclinations of which to the axes of coordinates are f, g, h, the values of λ, μ, ν then being $\tan\frac{1}{2}\vartheta\cos f$, $\tan\frac{1}{2}\vartheta\cos g$, $\tan\frac{1}{2}\vartheta\cos h$ respectively: $(\lambda_1, \mu_1, \nu_1)$ and $(\lambda_2, \mu_2, \nu_2)$ have of course the like significations; and then, if (λ, μ, ν) refer to the first rotation, and $(\lambda_1, \mu_1, \nu_1)$ to the second rotation, the values of $(\lambda_2, \mu_2, \nu_2)$ for the rotation compounded of these are taken to be[2]

$$\begin{aligned}\lambda_2 &= \lambda + \lambda_1 + \mu\nu_1 - \mu_1\nu,\\ \mu_2 &= \mu + \mu_1 + \nu\lambda_1 - \nu_1\lambda,\\ \nu_2 &= \nu + \nu_1 + \lambda\mu_1 - \lambda_1\mu,\end{aligned}$$

[2]The numerators might equally well have been $\lambda + \lambda_1 - (\mu\nu_1 - \mu_1\nu)$, etc., but there is no essential difference: we pass from one set of formulæ to the other by reversing the signs of all the symbols: and hence, by properly fixing the sense of the rotations, the signs may be made to be $+$ as in the text. Assuming this to be so, if we then reverse the order of the component rotations, we have for the new compound rotation the like formulæ with the signs $-$ instead of $+$; but this in passing. The formulæ, virtually due to Rodrigues, are given in my paper "On the motion of rotation of a solid body," *Camb. Math. Journal,* t. III. (1843), [6].

each divided by

$$1 - \lambda\lambda_1 - \mu\mu_1 - \nu\nu_1;$$

and if we then write for λ, μ, ν, the quotients x, y, z each divided by w, and in like manner for λ_1, μ_1, ν_1 and λ_2, μ_2, ν_2, the quotients x_1, y_1, z_1 each divided by w_1, and x_2, y_2, z_2 each divided by w_2, the formulae for the composition of the rotations are

$$\begin{aligned} x_2 &= xw_1 + x_1w + yz_1 - y_1z, \\ y_2 &= yw_1 + y_1w + zx_1 - z_1x, \\ z_2 &= zw_1 + z_1w + xy_1 - x_1y, \\ w_2 &= ww_1 - xx_1 - yy_1 - zz_1; \end{aligned}$$

and the question is to express A, B, C, D as functions of (x, y, z, w), such that A_1, B_1, C_1, D_1 denoting the like functions of (x_1, y_1, z_1, w_1), A_2, B_2, C_2, D_2 shall denote the like functions of (x_2, y_2, z_2, w_2).

It is found that the required conditions are satisfied by assuming

$$A, B, C, D = ix - y,\ -iz + w,\ -iz - w,\ -ix - y,$$

(where $i = \sqrt{-1}$ as usual): in fact, we then have

$$\begin{aligned} A_2 &= B_1A - A_1C \\ &= (-iz_1 + w_1)(ix - y) - (ix_1 - y_1)(-iz - w) \\ &= i(xw_1 + x_1w + yz_1 - y_1z) - (yw_1 + y_1w + zx_1 - z_1x) \end{aligned}$$

as it should be: and we verify in like manner the values of B_2, C_2 and D_2 respectively.

The result consequently is that we have the homography

$$(ix - y) + (-iz + w)p + (-iz - w)q + (-ix - y)pq = 0$$

corresponding to the rotation $\left(\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right)$: where $\frac{x}{w}, \frac{y}{w}, \frac{z}{w}$ are the parameters of rotation, $\tan\frac{1}{2}\vartheta\cos f$, $\tan\frac{1}{2}\vartheta\cos g$, $\tan\frac{1}{2}\vartheta\cos h$.

I remark as regards the geometrical theory that, if we consider two lines J and K fixed in space, and a third line L fixed in the solid body and moveable with it: then, for any given position of the solid body, the three lines J, K, L are directrices of a hyperboloid, the generatrices whereof meet each of the three lines: and these generatrices determine, say on the fixed lines J and K, two series of points corresponding homographically to each other: that is, corresponding to any given position of the solid body we have a homography. But it is not immediately obvious how we can thence obtain the foregoing analytical formula.

Cambridge, 3 *April,* 1879.

661.

ON THE DOUBLE ϑ-FUNCTIONS.

[From the *Proceedings of the London Mathematical Society*, vol. IX. (1878), pp. 29, 30.]

Prof. Cayley gave an account of researches[3] on which he is engaged upon the double ϑ-functions. In regard to these, he establishes in a strictly analogous manner the theory of the single ϑ-functions, the process for the single functions being in fact as follows: Considering u, x as connected by the differential relation

$$du = \frac{dx}{\sqrt{\overline{a-x}\,.\,\overline{b-x}\,.\,\overline{c-x}\,.\,\overline{d-x}}},$$

then, if A, B, C, D, Ω denote functions of u, viz. for shortness, the single letters are used, instead of writing them as functional symbols, $A(u), B(u)$, &c., then, by way of definition of these functions (called, the first four of them ϑ-functions, and the last an ω-function), we assume

$$A, B, C, D = \Omega\sqrt{a-x},\ \Omega\sqrt{b-x},\ \Omega\sqrt{c-x},\ \Omega\sqrt{d-x}$$

respectively, together with one other equation, as presently mentioned. Without in any wise defining the meaning of Ω, we then obtain a set of equations of the form

$$A\delta B - B\delta A = \&\text{c. } \sqrt{\overline{c-x}\,.\,\overline{d-x}}\,\delta u,$$

(mere constant coefficients are omitted), or, what is the same thing,

$$A\delta B - B\delta A = CD\,\delta u,$$

which are differential equations defining the nature of the ratio-functions $A : B : C : D$. If, proceeding to second differential coefficients, we attempt to form the expressions for $A\delta^2 A - (\delta A)^2$, &c., these involve multiples of $\Omega\delta^2\Omega - (\delta\Omega)^2$; in order to obtain a con-

[3]See paper, number 665.